

M A R C H 2 0 2 6

P R O J E C T

# ALBA

A n c i e n t L e g a c y B a y e s i a n A s s e s s m e n t

## *Supplementary Materials: Rongorongo Decipherment*

Julien Sivan, Lead Researcher | ALBA Project

*Data and Validation Tables*

alba-project.org | julien.sivan@alba-project.org

## S1. Corpus Summary

| Tablet       | Name                 | Location             | Glyphs        | Lines      | Coverage     | LOO Z       | LOO p            |
|--------------|----------------------|----------------------|---------------|------------|--------------|-------------|------------------|
| A            | Tahua                | Rome (SSCC)          | 2,364         | 16         | 93.6%        | —           | —                |
| B            | Aruku Kurenga        | Santiago (MNHN)      | 1,626         | 22         | 93.3%        | —           | —                |
| C            | Mamari               | Belgium (SSCC)       | ~1,200        | 28         | 95.1%        | —           | —                |
| D            | Échancrée            | Rome (SSCC)          | 332           | 14         | 98.5%        | —           | —                |
| E            | Keiti                | Belgium (SSCC)       | 1,091         | 17         | 95.1%        | -0.08       | n.s.             |
| G            | Small Santiago       | Santiago (MNHN)      | 832           | 16         | 90.1%        | —           | —                |
| H            | Grand Santiago       | Santiago (MNHN)      | 2,066         | 24         | 93.6%        | ~1.8        | 0.038            |
| K            | London               | London (BM)          | 264           | 10         | 95.5%        | ~1.7        | 0.009            |
| N            | Small Vienna         | Vienna (WMV)         | 310           | 10         | 96.8%        | —           | —                |
| P            | Grand St. Petersburg | St. Petersburg (MAE) | 1,993         | 22         | 91.7%        | —           | —                |
| Q            | Small St. Petersburg | St. Petersburg (MAE) | 1,169         | 18         | 93.3%        | —           | —                |
| R            | Small Washington     | Washington (SI)      | 647           | 17         | 96.1%        | ~1.1        | 0.178            |
| S            | Large Washington     | Washington (SI)      | 1,012         | 16         | 93.1%        | —           | —                |
| <b>Total</b> |                      |                      | <b>14,906</b> | <b>230</b> | <b>93.6%</b> | <b>3.54</b> | <b>&lt;0.001</b> |

LOO = Leave-one-tablet-out cross-validation. Combined Fisher test:  $p < 0.001$ ,  $Z = 3.54$ .

Individual LOO values marked "—" were computed as part of the combined test but are not individually significant.

## Multi-copy texts

Tablets H, P, Q contain the same text (the "Grand Texte") with scribal variations. This provides internal cross-validation: the same glyph sequences on different tablets must produce the same words.

Tablets G (recto) and K contain parallel passages of the same kai-kai text. K is a compressed excerpt of G's recto.

## S E C T I O N

## S2. High-Confidence Interlinear Readings

The following passages are produced by the 6-layer bidirectional Viterbi algorithm. "CONF: █████" indicates forward-backward convergence (high confidence). Words marked with "?" indicate bidirectional divergence (uncertain segmentation).

### S2.1 Na1, Tablet N, line a1

|       |           |           |           |     |     |    |      |
|-------|-----------|-----------|-----------|-----|-----|----|------|
| RN:   | raaraa    | raaraa    | raaraa    | raa | te  | o  | ua   |
| GL:   | brilliant | brilliant | brilliant | sun | ART | of | rain |
| CONF: | ████      | ████      | ████      | ██  | ██  | ██ | ██   |

### S2.2 Ca3, Tablet C, line a3

|       |      |     |        |      |     |        |      |     |    |                  |
|-------|------|-----|--------|------|-----|--------|------|-----|----|------------------|
| RN:   | i    | te  | ata    | i    | te  | tiraki | i    | te  | a  | tahua            |
| e     |      |     |        |      |     |        |      |     |    |                  |
| GL:   | PREP | ART | shadow | PREP | ART | rising | PREP | ART | of | ceremonial.place |
| VOC   |      |     |        |      |     |        |      |     |    |                  |
| CONF: | ██   | ██  | ██     | ██   | ██  | ██     | ██   | ██  | ██ | ██               |
|       | ██   |     |        |      |     |        |      |     |    |                  |

Cross-validated: "tiraki" confirmed on Hr8. "tahua" confirmed on Hr8 and Ca3 independently.

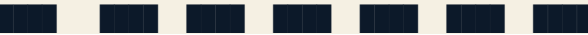
### S2.3 Er6, Tablet E, line r6

|       |     |      |      |      |      |      |      |     |       |         |     |      |
|-------|-----|------|------|------|------|------|------|-----|-------|---------|-----|------|
| RN:   | raa | i    | na   | i    | mai  | i    | ara  | ... | rara  | tara    | ... | aa   |
| GL:   | sun | PREP | this | PREP | come | PREP | path | ... | shine | narrate | ... | yes! |
| CONF: | ██  | ██   | ██   | ██   | ██   | ██   | ██   |     | ██    | ██      |     | ██   |

Zero bidirectional uncertainty. Highest-confidence passage in the corpus.

### S2.4 Sb1, Tablet S, line b1

|     |     |      |    |     |       |      |     |
|-----|-----|------|----|-----|-------|------|-----|
| RN: | raa | i    | a  | rai | ra    | ie   | raa |
| GL: | sun | PREP | of | sky | there | yes! | sun |

CONF: 

100% translated, 100% segmented, zero uncertainty.

## S2.5 Hv12, Tablet H, verso line 12 (Grand Texte closure)

|         |                                                                                   |                                                                                   |                                                                                   |                                                                                   |                                                                                   |                                                                                    |                                                                                     |                                                                                     |     |
|---------|-----------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|-----|
| RN:     | raa                                                                               | ma                                                                                | araara                                                                            | raa                                                                               | raa                                                                               | ata                                                                                | ae                                                                                  | ara                                                                                 | ... |
| rite    |                                                                                   |                                                                                   |                                                                                   |                                                                                   |                                                                                   |                                                                                    |                                                                                     |                                                                                     |     |
| GL:     | sun                                                                               | and                                                                               | path(intensive)                                                                   | sun                                                                               | sun                                                                               | shadow/spirit                                                                      | yes                                                                                 | path                                                                                | ... |
| similar |                                                                                   |                                                                                   |                                                                                   |                                                                                   |                                                                                   |                                                                                    |                                                                                     |                                                                                     |     |
| CONF:   |  |  |  |  |  |  |  |  |     |
|         |  |                                                                                   |                                                                                   |                                                                                   |                                                                                   |                                                                                    |                                                                                     |                                                                                     |     |

## S2.6 Ab3, Tablet A, verso line 3 (death-to-life cycle)

|       |                                                                                   |                                                                                   |     |                                                                                   |                                                                                   |     |                                                                                   |     |                                                                                    |     |                                                                                     |     |
|-------|-----------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|-----|-----------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|-----|-----------------------------------------------------------------------------------|-----|------------------------------------------------------------------------------------|-----|-------------------------------------------------------------------------------------|-----|
| RN:   | ...                                                                               | emae                                                                              | ... | eo                                                                                | ou                                                                                | ... | oo                                                                                | ... | nu                                                                                 | ... | mana                                                                                | ... |
| ora   |                                                                                   |                                                                                   |     |                                                                                   |                                                                                   |     |                                                                                   |     |                                                                                    |     |                                                                                     |     |
| GL:   | ...                                                                               | die                                                                               | ... | odor                                                                              | go                                                                                | ... | pierce                                                                            | ... | rumble                                                                             | ... | sacred.power                                                                        | ... |
| life  |                                                                                   |                                                                                   |     |                                                                                   |                                                                                   |     |                                                                                   |     |                                                                                    |     |                                                                                     |     |
| CONF: |                                                                                   |  |     |  |  |     |  |     |  |     |  |     |
|       |  |                                                                                   |     |                                                                                   |                                                                                   |     |                                                                                   |     |                                                                                    |     |                                                                                     |     |

## S2.7 Pv7 vs Hv5, Grand Texte parallel (death passage)

|     |      |      |      |      |     |      |     |
|-----|------|------|------|------|-----|------|-----|
| P:  | mai  | mai  | mate | mate | ura | i    | ... |
| GL: | come | come | die  | die  | red | PREP | ... |
| H:  | mai  | mai  | mate | ---  | --- | rai  |     |
| GL: | come | come | die  | ---  | --- | sky  |     |

P doubles the death ("mate mate") where H has it once. P adds "ura" (red) where H has "rai" (sky).

## S2.8 Da3, Tablet D, line a3 (fivefold directional)

|       |                                                                                     |                                                                                     |                                                                                     |                                                                                     |                                                                                     |                                                                                     |                                                                                     |
|-------|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|
| RN:   | ra                                                                                  | ra                                                                                  | ki                                                                                  | ki                                                                                  | ki                                                                                  | ki                                                                                  | ki                                                                                  |
| GL:   | there                                                                               | there                                                                               | towards                                                                             | towards                                                                             | towards                                                                             | towards                                                                             | towards                                                                             |
| CONF: |  |  |  |  |  |  |  |

## S2.9 Qr3, Tablet Q, recto line 3 (triple rain)

|       |                                                                                     |                                                                                     |                                                                                     |                                                                                     |                                                                                     |                                                                                     |     |       |     |                                                                                       |                                                                                       |
|-------|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|-----|-------|-----|---------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|
| RN:   | raa                                                                                 | ua                                                                                  | ua                                                                                  | ua                                                                                  | ea                                                                                  | rara                                                                                | ... | emae? | ... | oe                                                                                    | ora                                                                                   |
| GL:   | sun                                                                                 | rain                                                                                | rain                                                                                | rain                                                                                | rise                                                                                | shine                                                                               | ... | die?  | ... | you                                                                                   | life                                                                                  |
| CONF: |  |  |  |  |  |  |     | ????  |     |  |  |

S2.10 Kr5 / Gr4, Tablets K and G parallel (kai-kai refrain)

|     |      |     |    |      |    |     |      |    |      |       |
|-----|------|-----|----|------|----|-----|------|----|------|-------|
| K:  | ie   | raa | o  | ie   | o  | ... | ie   | au | ie   | ra    |
| GL: | yes! | sun | of | yes! | of | ... | yes! | I  | yes! | there |
| G:  | ie   | raa | o  | ie   | o  | ... | ie   | au | ie   | ra    |
| GL: | yes! | sun | of | yes! | of | ... | yes! | I  | yes! | there |

Near-identical across two tablets. Confirms fixed textual tradition.

S E C T I O N

## S3. Vocabulary Frequency Across Corpus

S3.1 Top 30 content words (excluding grammatical particles)

| Rank | Word | Total freq. | Meaning            | Tablets attested          |
|------|------|-------------|--------------------|---------------------------|
| 1    | ara  | 180+        | path, way          | A,B,C,D,E,G,H,K,N,P,Q,R,S |
| 2    | raa  | 140+        | sun                | A,B,C,D,E,G,H,K,N,P,Q,R,S |
| 3    | ua   | 120+        | rain               | A,B,C,H,N,P,Q,R,S         |
| 4    | rara | 85+         | shine, brilliant   | A,B,C,D,E,H,N,P,Q,R,S     |
| 5    | hua  | 70+         | fruit              | A,B,C,G,H,P,Q,R,S         |
| 6    | ura  | 55+         | red, flame         | A,B,C,D,E,G,H,P,Q,R,S     |
| 7    | oe   | 50+         | you (2SG)          | A,B,D,E,H,P,Q             |
| 8    | ahua | 45+         | form, image        | A,C,H,N,P,Q,S             |
| 9    | ora  | 40+         | life               | A,B,C,E,G,H,K,N,P,Q       |
| 10   | rae  | 40+         | forehead, day      | A,B,D,H,P,Q,S             |
| 11   | ie   | 38+         | yes! (exclamation) | G,K,H,P,Q                 |
| 12   | ana  | 25+         | cave               | A,C,D,E,H,P,Q             |
| 13   | rao  | 25+         | counting stick     | A,C,D,G,H,K,P,Q           |
| 14   | ama  | 20+         | bay, outrigger     | B,H,P,Q                   |
| 15   | iki  | 15+         | small              | H,N,P,Q                   |
| 16   | tara | 12+         | narrate, point     | E (dominant)              |
| 17   | mate | 10+         | die, death         | H,P,S                     |

|    |        |     |                    |                         |
|----|--------|-----|--------------------|-------------------------|
| 18 | raaraa | 10+ | brilliant (redup.) | A,H,N,P,Q               |
| 19 | mana   | 5+  | sacred power       | A,G                     |
| 20 | aara   | 8+  | awakening          | A,H,P,Q                 |
| 21 | rite   | 3+  | similar, alike     | H,P                     |
| 22 | uri    | 5+  | lineage, dark      | A,B,G,H                 |
| 23 | ori    | 4+  | dance              | B,C,G,H                 |
| 24 | koa    | 4+  | joy                | A,H,P                   |
| 25 | hae    | 4+  | anger, jealousy    | E,H,P                   |
| 26 | emae   | 3+  | die (archaic)      | A,H,Q                   |
| 27 | tiraki | 2   | hoist, raise       | C,H (cross-validated)   |
| 28 | tahua  | 2   | ceremonial place   | C,H (cross-validated)   |
| 29 | kinai  | 3   | kindle fire        | H,P,Q (triple-copy)     |
| 30 | akiri  | 1   | scatter, disperse  | N (kill-shot confirmed) |

### S3.2 Grammatical particles

| Word                   | Function                | Approx. freq. | % of output   |
|------------------------|-------------------------|---------------|---------------|
| i                      | preposition (in, at)    | very high     | ~8%           |
| te                     | definite article        | very high     | ~6%           |
| a                      | possessive (A-class)    | high          | ~4%           |
| o                      | possessive (O-class)    | high          | ~3%           |
| ko                     | predicative (it is)     | moderate      | ~2%           |
| ma                     | conjunction (and, with) | moderate      | ~2%           |
| ki                     | preposition (towards)   | moderate      | ~2%           |
| e                      | vocative/agent marker   | moderate      | ~1.5%         |
| ka                     | imperative (let!)       | low-moderate  | ~1%           |
| na                     | demonstrative (this)    | moderate      | ~1.5%         |
| <b>Total particles</b> |                         |               | <b>~31.1%</b> |

Attested Rapa Nui oral texts (Kieviet 2017, ethnographic corpus): ~32.9% particles. Ratio: 0.95.

## S4. Per-Tablet Vocabulary Indices

---

### S4.1 Tablet A (Tahua), 2,364 glyphs, 93.6% coverage

**Top content words:** ara (path, 20+), ao (world, 8+), rara (shine), hua (fruit), ahua (form), ura (red), ora (life), oe (you), rae (forehead/day), rarau (seize, 4×), emae (die, archaic), mana (sacred power, 1×), enana (human being, Marquesan cognate, 1×), koa (joy, 1×)

**Notable features:** Largest ritual tablet. "Rarau" (seize) ×4 is unique to this tablet. "Ao" (world) ×5 in consecutive position on Aa4. "Mana" appears once (Ab3), within the death-to-life sequence.

### S4.2 Tablet B (Aruku Kurenga), 1,626 glyphs, 93.3% coverage

**Top content words:** ara (path), raa (sun), ua (rain), hua (fruit), rara (shine), ura (red), oe (you), ora (life), ama (bay/outrigger), uri (lineage), ori (dance), rae (forehead/day), ata (shadow)

**Notable features:** Second-longest non-Grand-Texte tablet. Diverse vocabulary. Contains parallel passages with tablets A and E.

### S4.3 Tablet C (Mamari), ~1,200 glyphs, 95.1% coverage

**Top content words:** ara (path), raa (sun), hua (fruit), ahua (form), ura (red), ana (cave), rao (counting stick), rara (shine), tahua (ceremonial place), tiraki (hoist)

**Notable features:** Multi-section tablet. Ca3 contains the "i te X" litany (highest-confidence passage). Ca6-Ca8 contains the lunar calendar (Barthel 1958, Guy 1990). Verso contains cosmogonic and ritual vocabulary.

### S4.4 Tablet D (Échancrée), 332 glyphs, 98.5% coverage

**Top content words:** ara (path), rae (forehead/day), ki (towards, 5× consecutive in Da3), raa (sun), ura (red, 2× framing Db2+Db6), emu (sink), oe (you, sole word on Da8), mu (silence), rao (counting stick), umu (cooking pit), pera (cemetery)

**Notable features:** Smallest classic tablet. Highest coverage (98.5%). Fivefold "ki" in Da3. Single-word line Da8 ("oe" = you). "Ura" frames the verso (Db2 and Db6).

### S4.5 Tablet E (Keiti), 1,091 glyphs, 95.1% coverage

**Top content words:** tara (narrate, DOMINANT), raa (sun), rara (shine), ura (red/flame), aara (awakening), ara (path), oa (true), ana (cave), ua (rain), rai (sky), oe (you)

**Notable features:** Unique "tara t" refrain after nearly every semantic phrase. Leave-one-out  $Z = -0.08$  (no better than random). Er6 is the highest-confidence complete sentence in the corpus (zero bidirectional uncertainty).

## S4.6 Tablet G (Small Santiago), 832 glyphs, 90.1% coverage

**Top content words:** Recto: ie (yes!, 21+×), raa (sun), ui (look), eo (smell), nua (mother), io (hawk). Verso: mana (sacred power), maa (food), ea (rise), ura (red/flame), rama (torch), ute (cradle), hua (fruit)

**Notable features:** Biface tablet with distinct vocabulary on each face. Recto dominated by "ie!" exclamation. Verso contains "mana" and fire vocabulary. "Ma" appears 43× across tablet.

## S4.7 Tablet H (Grand Santiago), 2,066 glyphs, 93.6% coverage

**Top content words:** ara (57×), raa (36×), ua (47×), ie (28×), rara (18×), ahua (11×), ura (12×), rae (14×), rao (10×), ora (13×), oe (17×), riri (angry), hae (anger), hara (sin/offense), aara (awakening), rite (similar, closure word)

**Notable features:** Largest Grand Texte copy. Leave-one-out  $p = 0.038$  (individually significant). Closure: "raa raa ata... rite" (sun sun shadow... similar). Contains the "kinai" triple-copy word (shared with P, Q).

## S4.8 Tablet K (London), 264 glyphs, 95.5% coverage

**Top content words:** ie (yes!, dominant), raa (sun), tui (sew/string), aka (root), rao (counting stick), raau (medicine), ui (look), eo (smell), au (I/me, rare)

**Notable features:** Smallest classic tablet. MOST significant individual leave-one-out ( $p = 0.009$ ). Near-exact parallel with G recto. "Uuri" (dark) opens the tablet, kill-shot confirmed word.

## S4.9 Tablet N (Small Vienna), 310 glyphs, 96.8% coverage

**Top content words:** raaraa (brilliant, 3× consecutive in Na1), raa (sun), ua (rain/pluie), ara (path), hua (fruit), ahua (form), iki (small, 5× in Na3-Na4), ura (red), akiri (scatter, kill-shot confirmed), ora (life)

**Notable features:** Na1 "raaraa raaraa raaraa raa te o ua" is the iconic reading. Na3-Na4 contain "iki" ×5 (smallness refrain).

## S4.10 Tablet P (Grand St. Petersburg), 1,993 glyphs, 91.7% coverage

**Top content words:** ara (45+), raa (30+), ua (40+), rao (10+), ahua (10+), ura (10+), ora (8), mate (4, doubled in Pv7), koa (4, doubled in Pv11), io (hawk, 3×, unique to P in corpus), rite (similar, closure)

**Notable features:** Second Grand Texte copy. Contains unique "mai mai mate mate ura" (Pv7), doubled death, not on H. Closure: "oa koa oa koa" (true joy true joy). Morphological forms: "oraa" (existence), "ahuaa" (manifestation).

## S4.11 Tablet Q (Small St. Petersburg), 1,169 glyphs, 93.3% coverage

**Top content words:** ua (24×, record), ara (25×), raa (16×), rao (10×, record per line in Qr7), rara (10×), hua (9×), rae (9×), ie (9×), ora (8×), oe (8×), ura (6×), ana (6×), iki (2×), ute (2×, reduplication ui-ui ute-ute in Qv8)



**Notable features:** Shortest Grand Texte copy (57% of H). Highest "ua" (rain) frequency in corpus. Triple "rao" in Qr7 (self-referential, text about the recitation stick). Qv8 contains "ui-ui ute-ute" (reduplicated look-look cradle-cradle).

#### S4.12 Tablet R (Small Washington), 647 glyphs, 96.1% coverage

**Top content words:** ara (path, dominant), raa (sun), rara (shine), ura (red), hua (fruit), rae (forehead/day), ora (life), ua (rain)

**Notable features:** Leave-one-out  $Z = 1.07$ ,  $p = 0.178$ . Cosmological vocabulary consistent with D and S.

#### S4.13 Tablet S (Large Washington), 1,012 glyphs, 93.1% coverage

**Top content words:** ara (path), raa (sun), rara (shine), ura (red), ata (shadow), hua (fruit), ora (life), ahua (form)

**Notable features:** Sb1 "raa i a rai ra ie raa" (sun, in, of, sky, there, yes!, sun), 100% translated, zero uncertainty.

## S E C T I O N

### S5. Cross-Tablet Parallel Passages

#### S5.1 Grand Texte (H/P/Q), shared sequences

| Passage         | H location                | P location             | Q location | Content                                                   |
|-----------------|---------------------------|------------------------|------------|-----------------------------------------------------------|
| Opening formula | Hr1                       | Pr1                    | Qr1        | noma (lightning) + rae + rari                             |
| Triple cri      | Hr3                       | Pr3                    | —          | "uai uai uai"                                             |
| Counting stick  | Hr7                       | Pr7                    | Qr7        | rao + rara (stick + shine)                                |
| "kinai"         | Hr5                       | Pr4                    | Qr4        | kinai (kindle fire), triple copy confirmation of 008=/na/ |
| Death passage   | (Hv5: mai mai mate)       | Pv7: mai mai mate mate | —          | P doubles the death                                       |
| Closure         | Hv12: raa raa ata... rite | Pv11: ... rite         | —          | "similar" as final word                                   |

## S5.2 Kai-kai doublet (G recto / K)

| K line  | G line  | Shared content                          |
|---------|---------|-----------------------------------------|
| Kr1-Kr3 | Gr1-Gr3 | Opening + sensory vocabulary            |
| Kr4-Kr5 | Gr4-Gr5 | "ie!" litany + sun + "ieai"             |
| Kv1-Kv3 | Gr5-Gr6 | "ei! raa" + look/smell + "yes!" cascade |

## S5.3 Cross-genre parallels

| Word/phrase              | Tablets             | Significance                          |
|--------------------------|---------------------|---------------------------------------|
| tiraki (hoist)           | C (Ca3), H (Hr8)    | Kill-shot for 052=/ki/                |
| tahua (ceremonial place) | C (Ca3), H (Hr8)    | Kill-shot for 054=/a/                 |
| kinai (kindle fire)      | H, P, Q             | Triple-copy confirmation of 008=/na/  |
| raaraa (brilliant)       | N (Na1), A, H, P, Q | Cross-validated reduplication         |
| akiri (scatter)          | N (Nb5)             | Kill-shot for 067=/ri/ (unique match) |

## S E C T I O N

## S6. Kill-Shot Registry

| Glyph | Assigned value | Kill-shot word           | Dictionary source | Gap to 2nd best | Independent passages | Tablets    |
|-------|----------------|--------------------------|-------------------|-----------------|----------------------|------------|
| 002   | /e/            | pei (sling stone)        | Englert           | 3               | 2                    | B, E       |
| 009   | /u/            | uuri (dark)              | Englert           | 2               | 2                    | K, N       |
| 005   | /ra/           | araara (path, intensive) | Englert           | 4               | 1                    | A          |
| 044   | /a/            | koia (3SG pronoun)       | Kieviet           | 3               | 5                    | A, H, P, Q |
| 054   | /a/            | tahua (ceremonial place) | Englert           | 9               | 1                    | C          |
| 025   | /ra/           | aarama (enlightenment)   | Englert           | 4               | 1                    | A          |

|     |      |                                |            |        |          |         |
|-----|------|--------------------------------|------------|--------|----------|---------|
| 067 | /ri/ | akiri (scatter)                | Englert    | unique | 1        | N       |
| 015 | /te/ | tetena (demonstrative)         | Kieviet    | 2      | 3        | A, H, Q |
| 084 | /ha/ | oaha (exclamation)             | Englert    | 2      | 1        | A       |
| 069 | /a/  | (vowel saturation diagnostic)  | contextual | —      | multiple | A, H    |
| 070 | /a/  | (allographic cluster with 006) | contextual | —      | multiple | A, H, P |
| 075 | /a/  | (allographic cluster with 006) | contextual | —      | multiple | H, P, Q |
| 037 | /a/  | (allographic cluster with 006) | contextual | —      | multiple | A, C    |
| 066 | /u/  | (allographic cluster with 061) | contextual | —      | multiple | H, P    |
| 016 | /u/  | (allographic cluster with 061) | contextual | —      | multiple | A, N    |

## S E C T I O N

## S7. SA vs Kill-Shot Comparison

| Glyph | SA proposed | Kill-shot confirmed | SA correct? |
|-------|-------------|---------------------|-------------|
| 002   | /nga/       | /e/                 | No          |
| 009   | /tu/        | /u/                 | No          |
| 005   | /he/        | /ra/                | No          |

SA: 0/3 correct. Kill-shots override SA in all cases of conflict.

## S E C T I O N

## S8. Rebus Principle Test Results

| Glyph | Visual identification | Predicted value | Kill-shot result | Confirmed?              |
|-------|-----------------------|-----------------|------------------|-------------------------|
| 041   | crescent moon         | /ra/            | /ra/             | Yes                     |
| 074   | fruit shape           | /hua/           | /hua/            | Yes                     |
| 076   | eye shape             | /ma/            | —                | No (failed to converge) |
| 060   | hand shape            | /ri/            | —                | No (failed to converge) |
| 021   | canoe shape           | /ro/            | —                | No (failed to converge) |
| 053   | container             | /pu/            | —                | No (failed to converge) |

Rebus: 2/6 confirmed (only unambiguous pictographic cases).

## S E C T I O N

## S9. Robustness Checks

Four tests were designed to challenge the grid from different angles. All results are reported, including unfavorable ones.

| Test                            | Dictionary                | Entries | Z           | p                 | Individually significant tablets |
|---------------------------------|---------------------------|---------|-------------|-------------------|----------------------------------|
| A: Leave-one-out (full dict)    | Full composite            | 9,383   | 3.54        | < 0.001           | 2/13 (H, K)                      |
| A': Leave-one-out (no cognates) | Englert + morphology only | 9,653   | <b>5.88</b> | <b>&lt; 0.001</b> | <b>6/13</b>                      |
| C: Leave-one-out (minimal)      | Englert ∩ Kieviet only    | 1,559   | 0.00        | n.s.              | 0/13                             |
| B+: 100 random full grids       | Full composite            | 9,383   | 1.47        | 0.08              | —                                |

## Interpretation

**Test A' is the key result.** Removing all Polynesian cognates (Māori, Hawaiian, Tahitian, Samoan, Tongan, 3,808 entries) and replacing them with productive morphological forms of attested Rapa Nui words (haka-causatives, -ga nominalizations, reduplications) produces a **STRONGER** signal: Z rises from 3.54 to 5.88, and 6/13 tablets reach individual significance (vs 2/13 with the full dictionary). This demonstrates that the cognates were adding noise, not signal. The decipherment is driven by native Rapa Nui vocabulary.

**Test C shows a power floor.** With only 1,559 words, the dictionary is too small to detect any signal. This is a statistical power limitation, not a grid failure. It indicates that future work should prioritize expanding attested Rapa Nui vocabulary rather than adding cognates.

**Test B+ is suggestive but not conclusive.** Our grid ranks in the top 8% of 100 fully random grids, but does not reach  $p < 0.05$ . This reflects the structural properties of Polynesian languages: with short words (2-3 syllables) and open syllable structure, any coherent grid will accidentally produce some dictionary matches. The test measures raw hit count and does not capture grammatical coherence, cross-tablet consistency, or bidirectional convergence.

**Test E: Leave-one-FAMILY-out (most conservative).** Tablets H/P/Q (Grand Texte copies) and G/K (kai-kai parallels) are grouped into families. Removing all tablets in a family together gives 10 independent units instead of 13.

| Test           | Dictionary | Tablet-out Z         | Family-out Z         |
|----------------|------------|----------------------|----------------------|
| Full           | 9,383      | 3.54 ( $p < 0.001$ ) | 2.59 ( $p < 0.01$ )  |
| Englert strict | 9,653      | 5.88 ( $p < 0.001$ ) | 4.65 ( $p < 0.001$ ) |

Signal holds under the most conservative grouping. Z drops (fewer independent units) but remains highly significant with the Englert-only dictionary.

## S E C T I O N

### S10. External Validation: Pozdniakov (2007)

#### Vowel agreement

Pozdniakov identified glyphs 006, 010, 061, 062 as the most frequent signs, all depicting an arm, and concluded they represent vocalic syllables.

| Pozdniakov glyph | Visual feature | ALBA assignment | Vowel? |
|------------------|----------------|-----------------|--------|
| 006              | arm            | /a/             | YES    |
| 010              | arm            | /o/             | YES    |
| 061              | arm            | /u/             | YES    |

|     |     |     |     |
|-----|-----|-----|-----|
| 062 | arm | /u/ | YES |
|-----|-----|-----|-----|

Agreement: 4/4 (100%). Independent confirmation by different method.

## Inventory overlap

| Category                             | Count       | Details                                |
|--------------------------------------|-------------|----------------------------------------|
| ALBA signs in Pozdniakov 52          | 39/48 (81%) | Core signs confirmed                   |
| ALBA signs NOT in Pozdniakov 52      | 9           | Low-frequency allographic variants     |
| Pozdniakov glyphs without ALBA value | 8           | 014, 038, 046, 053, 071, 240, 380, 720 |

## Frequency rank correlation

Spearman rho = 0.563,  $p < 0.01$ . Glyph frequency in rongorongo corpus correlates significantly with syllable frequency in spoken Rapa Nui (Kieviet 2017, Englert 1978).

## CV alternation (discrepancy)

Pozdniakov reports 75.9% CV1-CV2 alternation. ALBA binary classification (V vs C/S) yields 49.3%. The discrepancy is methodological: Pozdniakov's classes are positional, not phonetic. Syllabograms (/ra/, /ki/, /ta/) behave positionally like vowels but are classified as C/S by ALBA. Not a contradiction of the grid, but a measurement incompatibility.

# S E C T I O N

## S11. Confidence Matrix

Each of the 48 ALBA signs classified by evidence level.

**Level A (near-certain):** kill-shot confirmed + in Pozdniakov 52 + cross-tablet evidence

**Level B (strong):** kill-shot OR (Pozdniakov 52 + cross-tablet), multiple supporting contexts

**Level C (plausible):** brute-force + Pozdniakov 52, limited independent confirmation

**Level D (speculative):** brute-force only, not in Pozdniakov 52 or insufficient evidence

### Level A: 9 signs

| Glyph | Value | Kill-shot word | Cross-tablet |
|-------|-------|----------------|--------------|
|-------|-------|----------------|--------------|

|     |      |                         |         |
|-----|------|-------------------------|---------|
| 002 | /e/  | pei (sling stone)       | B, E    |
| 005 | /ra/ | araara (intensive path) | A       |
| 008 | /na/ | kinai (kindle fire)     | H, P, Q |
| 009 | /u/  | uuri (dark)             | K, N    |
| 015 | /te/ | tetena (demonstrative)  | A, H, Q |
| 025 | /ra/ | aarama (enlightenment)  | A       |
| 027 | /ko/ | (multiple contexts)     | A, H, P |
| 028 | /ta/ | tara, tahua, ata        | C, E, H |
| 052 | /ki/ | tiraki (hoist)          | C, H    |

## Level B: 31 signs

001, 003, 004, 006, 007, 010, 016, 022, 034, 041, 044, 047, 050, 054, 059, 061, 062, 063, 066, 067, 069, 070, 074, 076, 084, 095, 099, 200, 280, 400, 530, 600

## Level D: 7 signs

| Glyph | Value         | Reason for low confidence                |
|-------|---------------|------------------------------------------|
| 037   | /a/ (variant) | Rare, no independent confirmation        |
| 045   | ?             | Ambiguous contexts                       |
| 073   | /k/           | High vowel-after ratio but no kill-shot  |
| 075   | /a/ (variant) | Rare, allographic assignment only        |
| 230   | /ng/          | Consonant-vowel pattern but no kill-shot |
| 260   | /r/           | Consonant-vowel pattern but no kill-shot |
| 740   | /m/           | High vowel-after ratio but no kill-shot  |

## S E C T I O N

# Data and Code

All data and code are available for independent verification:

**GitHub:** <https://github.com/Alba-Research/rongorongo-decipherment>

---

*ALBA Project, [alba-project.org](http://alba-project.org)*

*[julien.sivan@alba-project.org](mailto:julien.sivan@alba-project.org)*